

Using the Resilience plug-in

This plug-in to EwE version 6.5 and newer computes resilience (Arreguín-Sánchez 2014) when Ecosim runs, allows users to explore this data in a graph, and saves this data to CSV file when explicitly activated.

Configuration

The plug-in offers three screens or forms to access its functionality: a supply / demand form, a resilience form, and an acknowledgements form. All three forms are accessible from the EwE Navigator master node “Ecosim > Output > Resilience” (Figure 1).

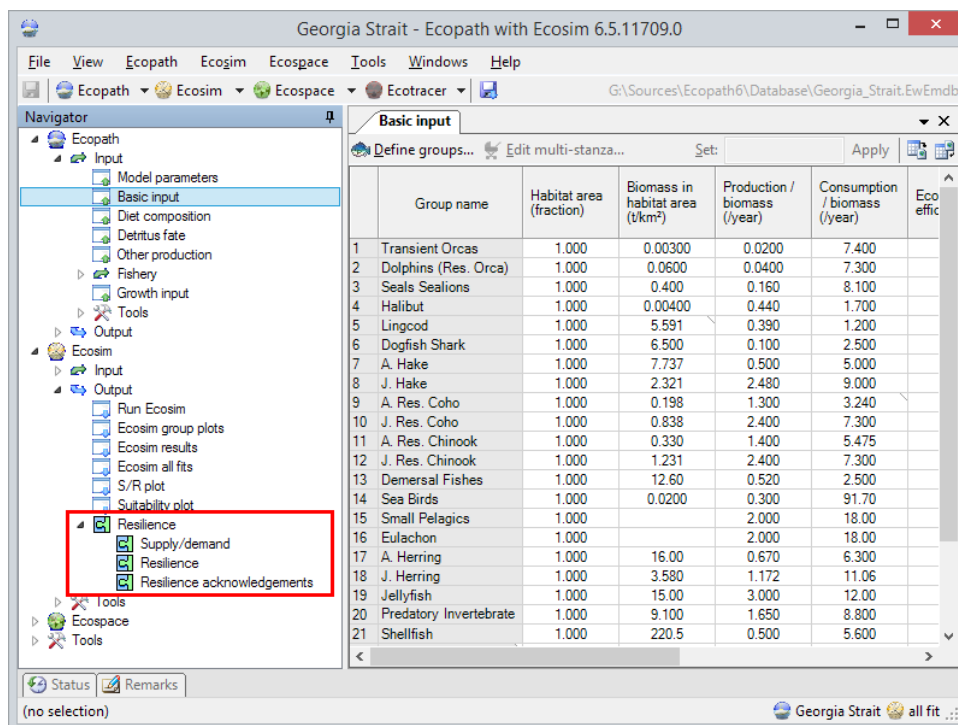


Figure 1 - Resilience navigation tree nodes

Supply and demand

The supply / demand form shows the logarithmic values of supply of all producers, and demand of all consumers over time (either annually or monthly), and shows a regression line that portrays resilience.

To show the supply / demand form, navigate to Ecosim > Output > Resilience > Supply / demand (Figure 2). Note that the plug-in will run Ecosim if this has not already been done.

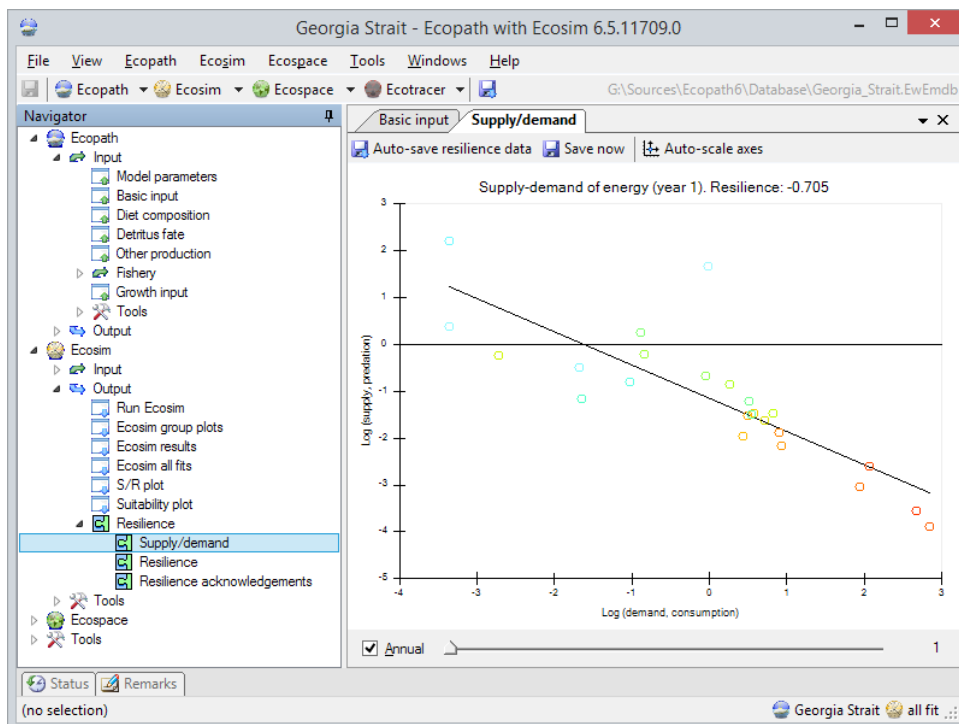


Figure 2 – The supply / demand form

The graph in this form displays monthly or annual resilience trends for the different consumers in the Ecopath model. The time slider can be used to display resilience for different Ecosim time steps (when showing monthly data) or years (when showing annual data).

To change between annual and monthly mode, check or uncheck the “Annual” check box.

To see values for a different time move the slider knob to a different value. Note that the current time (either year or Ecosim time step) is reflected to the right of the time slider.

The form offers three additional options:

- When the auto-save option is clicked, the plug-in will automatically save computed data to the default Ecosim output location. Saving happens after an Ecosim run. The Ecosim file location can be configured in the EwE File Management Options interface (see “Auto-saving”, below);
- When the “save now” option is clicked, the plug-in will save computed data immediately to the default Ecosim output location;
- When the “Auto-scale” option is checked, the graph will scale the X and Y axes best fit the computed data for each time step: when sliding across time, axes ranges will vary. When the option is not checked, the graph will scale its X and Y axes to the maximum supply and demand values encountered across the Ecosim run: when sliding across time, axes ranges will remain constant.

Resilience

The resilience form shows computed resilience for all Ecosim time steps and Ecosim years.

To show the resilience form, navigate to Ecosim > Output > Resilience > Resilience (Figure 3). Note that the plug-in will run Ecosim if this has not already been done.

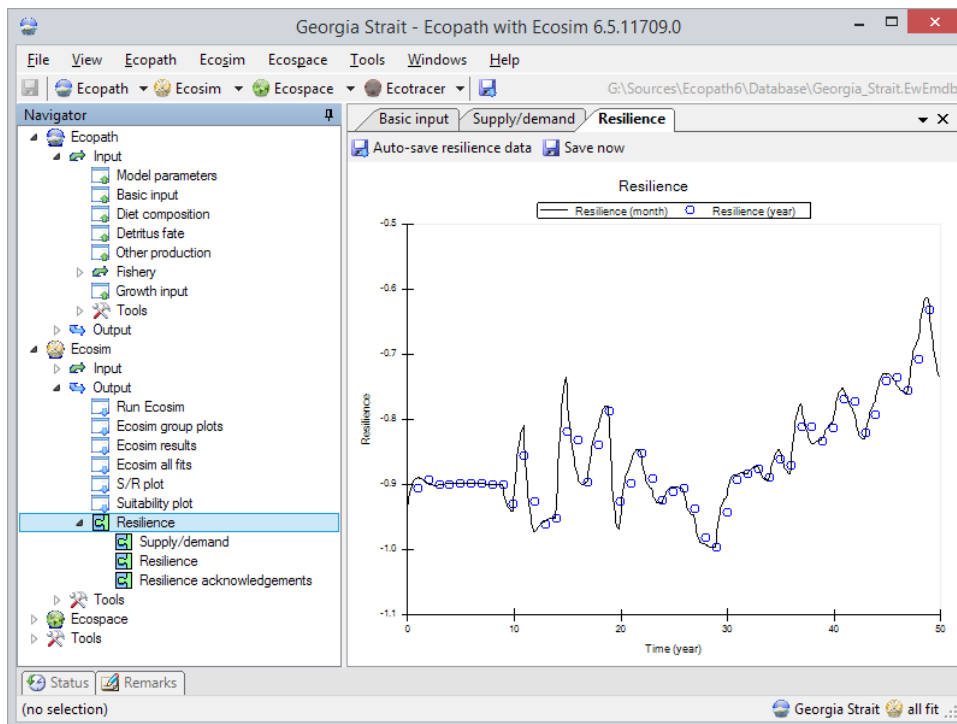


Figure 3 - The resilience form

The graph in this form displays both monthly and annual resilience trends for the Ecosim run. Time step values are depicted by a continuous line. Annual averages are depicted as blue circles.

The form offers two additional options:

- When the auto-save option is clicked, the plug-in will automatically save computed data to the default Ecosim output location. Saving happens after an Ecosim run. The Ecosim file location can be configured in the EwE File Management Options interface (see “Auto-saving”, below);
- When the “save now” option is clicked, the plug-in will save computed data immediately to the default Ecosim output location;

Acknowledgements

The acknowledgements form shows credits, references, sponsor information and a support link for the plug-in.

To show the acknowledgements form, navigate to Ecosim > Output > Resilience > Resilience acknowledgements (Figure 4).

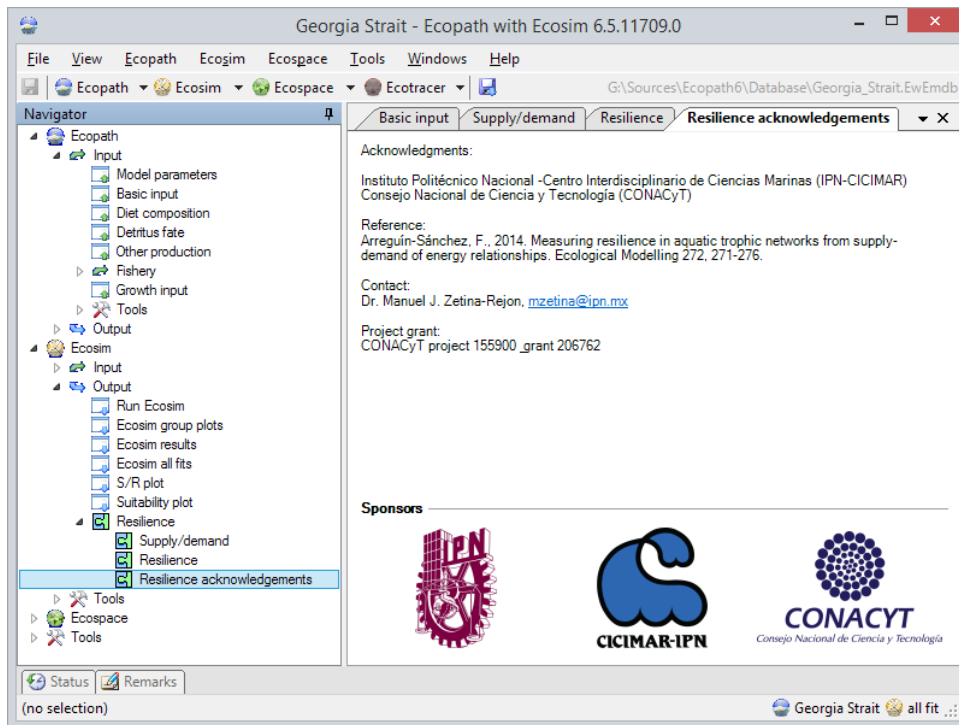


Figure 4 - The acknowledgements form

Clicking the logos will direct users to the website of each institute.

Clicking the support email address will attempt to launch the default mail client on the Windows machine, and will attempt to start a new email to the contact address.

Auto-saving

The Resilience can be set to automatically save its data to file. This option can also be controlled from the EwE file management settings (Tools > Options > File Management, Figure 2), where users can also configure where Ecosim will write its data to.

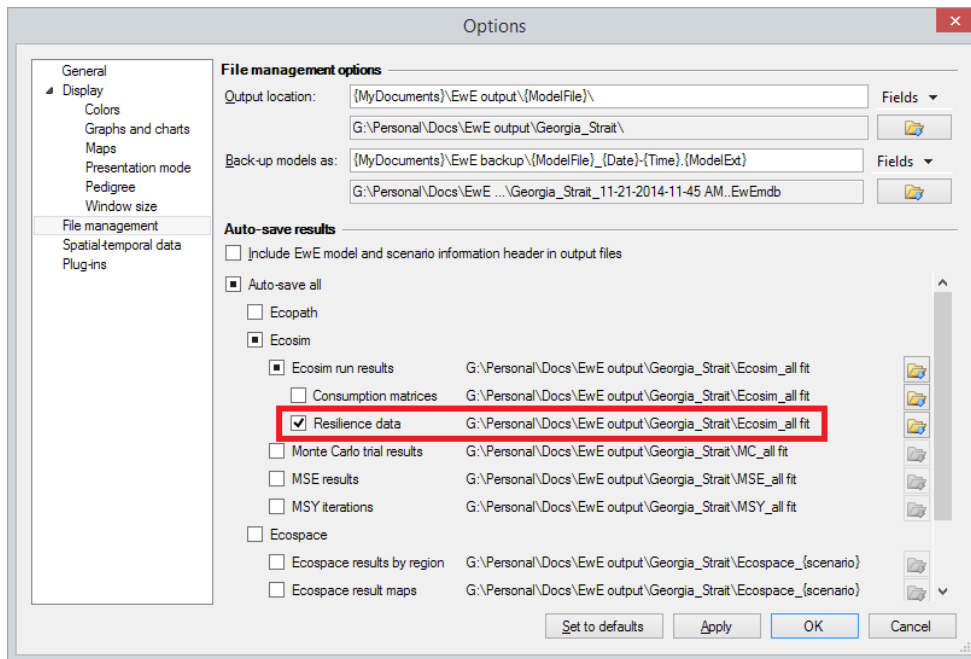


Figure 5 - Activating the resilience auto-save option

Whenever this option is active during an Ecosim run resilience CSV files will be saved to the path indicated in the auto-save settings for every time step, and for annual average values. The EwE Status panel will inform the user that resilience data has been saved, and where (Figure 3). Underlined nodes in this panel can be clicked to open the folder that data was saved to.

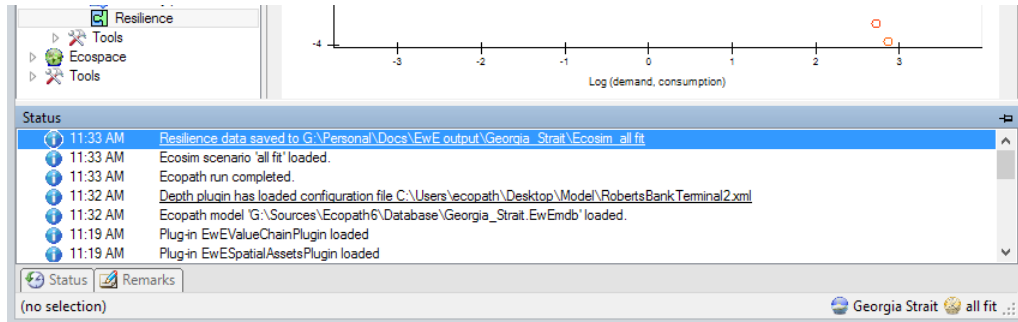


Figure 6 - The status panel provides information where data has been saved

CSV files will be named “Resilience_monthly.csv” and “Resilience_annual.csv”. CSV files are formatted just like any other CSV produced by Ecosim.

Details to be aware of

The plug-in adheres to EwE-wide export settings:

- Numbers will be formatted according to US English notations, with decimal points and without thousands separators;
- CSV files will be comma separated;
- CSV files will include the EwE model and scenario header unless this functionality is explicitly turned off in the auto-save settings (see Figure 1)

References

Arreguín-Sánchez, F. (2014). Measuring resilience in aquatic trophic networks from supply–demand–of-energy relationships. *Ecological Modelling* 272, 271–276.